آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 2

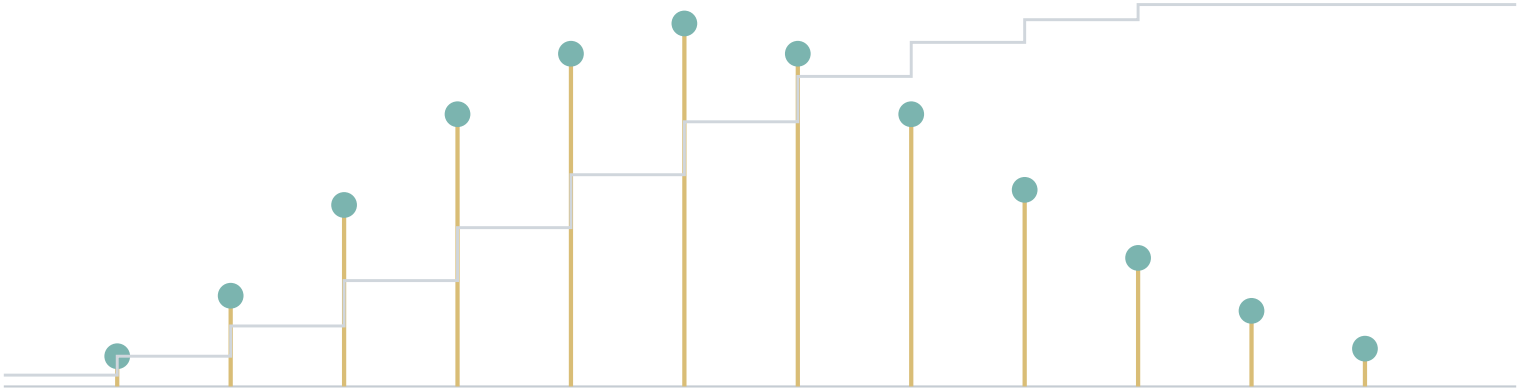
One Random Variable and Its Functions

یک متغیر تصادفی و توابع آن

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Learning Objectives

After completing this module, students will be able to:

1. Define random variables and explain their role in engineering analysis
2. Distinguish between discrete and continuous random variables
3. Compute and interpret PMF, PDF, and CDF
4. Clearly explain the difference between probability mass, density, and cumulative probability
5. Apply transformations to random variables
6. Analyze and visualize random variables

2.1 Introduction: Why Random Variables?

The Problem

In Module 1, we worked with events — subsets of a sample space. But engineering analysis requires **numbers**:

- How much noise voltage is present?
- How many packets arrived?
- What is the received signal power?

A **random variable** maps outcomes of a random experiment to real numbers, enabling us to use all the tools of calculus and algebra.

Engineering Motivation

We want to analyze…	We define a random variable…
Noise in a circuit	X = noise voltage (volts)
Bit errors in a frame	N = number of bit errors
Component lifetime	T = time until failure (hours)
Signal amplitude	A = received amplitude (V)
Packet delay	D = end-to-end delay (ms)

2.2 Definition of a Random Variable

Formal Definition

A **random variable** X is a function that maps each outcome ω in the sample space S to a real number:

$$X : S \to \mathbb{R}$$

$$\omega \mapsto X(\omega)$$

Interpretation

The random variable assigns a numerical value to each possible outcome of the experiment. The randomness comes from which outcome ω occurs — once ω is determined, $X(\omega)$ is a definite number.

Engineering Example

Experiment: Transmit 5 bits over a noisy channel.
Sample space: S = all possible sequences of correct/error for 5 bits ($2^5 = 32$ outcomes)
Random variable: X = number of bit errors

Outcome ω	$X(\omega)$
(C, C, C, C, C)	0
(E, C, C, C, C)	1
(E, E, C, C, C)	2
…	…
(E, E, E, E, E)	5

Now instead of working with 32 outcomes, we work with a single number $X \in \{0, 1, 2, 3, 4, 5\}$.

2.3 Discrete Random Variables

Definition

A random variable X is **discrete** if it takes values from a **countable** set (finite or countably infinite).

Characteristics

- Values can be listed: $\{x_1, x_2, x_3, \ldots\}$
- Each value has a specific probability (a "mass" of probability)
- Sum of all probabilities = 1

Engineering Examples

Random Variable	Possible Values	Context
Number of bit errors	$\{0, 1, 2, \ldots, n\}$	Packet transmission
Number of retransmissions	$\{0, 1, 2, \ldots\}$	ARQ protocol
Quantized voltage level	$\{0, 1, \ldots, 2^N - 1\}$	N-bit ADC output
Number of arrivals per second	$\{0, 1, 2, \ldots\}$	Network traffic
Number of defective chips	$\{0, 1, \ldots, n\}$	Wafer testing

2.4 Continuous Random Variables

Definition

A random variable X is **continuous** if it takes values from an **uncountable** set (interval of real numbers).

Characteristics

- X can take any value in a range (e.g., $[0, \infty)$ or $(-\infty, \infty)$)
- Probability of any single exact value is ZERO: $P(X = x) = 0$
- Probabilities are defined for intervals: $P(a \leq X \leq b)$

Engineering Examples

Random Variable	Range	Context
Thermal noise voltage	$(-\infty, \infty)$	Circuit noise
Time until failure	$[0, \infty)$	Reliability
Phase of received signal	$[0, 2\pi)$	Communications
Resistor value	$(R_0 - \delta, R_0 + \delta)$	Manufacturing
Channel gain	$[0, \infty)$	Wireless fading

Why $P(X = x) = 0$ for Continuous RVs

Imagine measuring a voltage. The probability that V = exactly 3.14159265… volts (infinite precision) is zero. We can only ask about intervals: $P(3.14 < V < 3.15)$.

2.5 Probability Mass Function (PMF)

Definition

For a discrete random variable X, the **probability mass function** is:

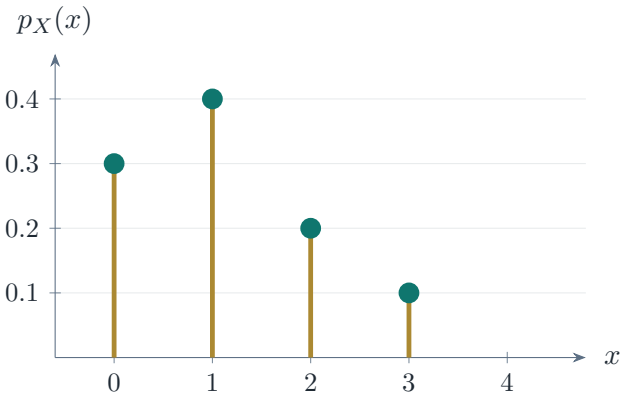
$$p_X(x) = P(X = x)$$

Properties

1. **Non-negativity:** $p_X(x) \geq 0$ for all x
2. **Normalization:** $\sum p_X(x_i) = 1$ (sum over all possible values)
3. **Probability computation:** $P(X \in A) = \sum_{x \in A} p_X(x)$

Visualization

The PMF is represented as **vertical lines** (impulses) at each possible value, with height equal to the probability.



Engineering Example: Bit Errors in a Byte

Transmit 8 bits, each with independent error probability $p = 0.1$.
X = number of bit errors.

$p_X(k) = C(8,\, k) \cdot (0.1)^k \cdot (0.9)^{8-k}$, for $k = 0,\, 1,\, \ldots,\, 8$

k	$p_X(k)$
0	0.4305
1	0.3826
2	0.1488
3	0.0331
4	0.0046
5+	< 0.001

2.6 Probability Density Function (PDF)

Definition

For a continuous random variable X, the **probability density function** $f_X(x)$ is defined such that:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

Properties

1. **Non-negativity:** $f_X(x) \geq 0$ for all x
2. **Normalization:** $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
3. **Probability = Area under curve** between two points

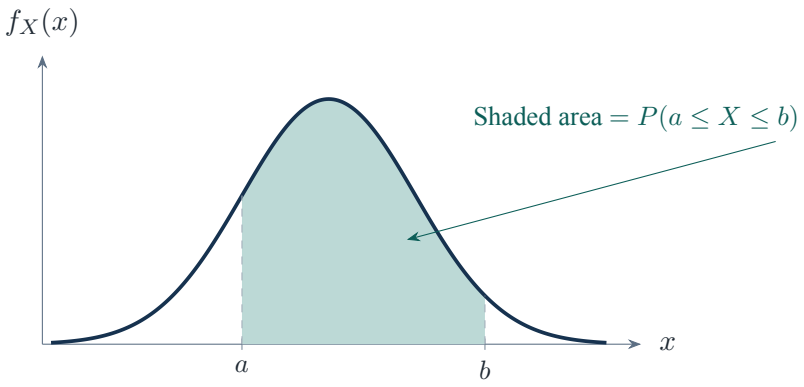
Critical Point: $f_X(x)$ is NOT a Probability!

⚠ $f_X(x)$ can be GREATER than 1!

The PDF gives probability **density** — probability per unit length. Only the **integral** (area) gives probability.

Analogy: Mass density (kg/m^3) is not mass. You need to integrate density over volume to get mass. Similarly, probability density is not probability — you integrate over an interval to get probability.

Visualization



Total area under curve $= 1$

Engineering Example: Thermal Noise Voltage

Thermal noise voltage in a circuit often follows a Gaussian distribution:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}$$

where σ is the RMS noise voltage.

$P(\text{noise exceeds } 2\sigma) = P(|X| > 2\sigma) = 1 - P(-2\sigma \leq X \leq 2\sigma) \approx 1 - 0.9545 = 0.0455$

2.7 Cumulative Distribution Function (CDF)

Definition

For any random variable X (discrete or continuous), the **cumulative distribution function** is:

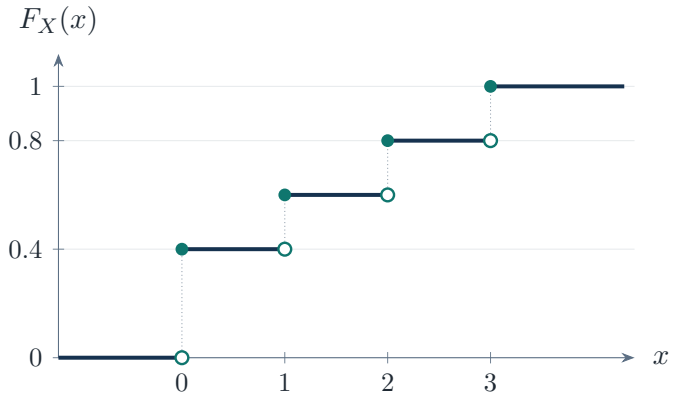
$$F_X(x) = P(X \leq x)$$

Properties

1. **Non-decreasing:** if $a < b$, then $F_X(a) \leq F_X(b)$
2. **Limits:** $F_X(-\infty) = 0$, $F_X(+\infty) = 1$
3. **Right-continuous:** $F_X(x)$ is continuous from the right
4. **Probability of interval:** $P(a < X \leq b) = F_X(b) - F_X(a)$

CDF for Discrete RVs

The CDF is a **staircase function** with jumps at each possible value:



Jump at $x = k$ has height $p_X(k)$.

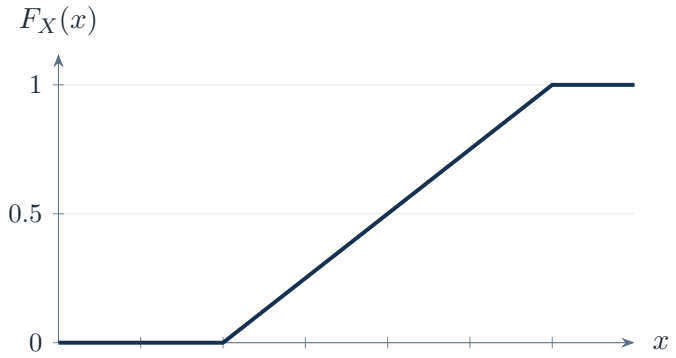
CDF for Continuous RVs

The CDF is a **smooth, continuous** curve:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

And conversely:

$$f_X(x) = \frac{dF_X(x)}{dx}$$



Relationship: PDF ↔ CDF

Direction	Formula	Meaning
PDF → CDF	$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$	Integrate density to get cumulative probability
CDF → PDF	$f_X(x) = \frac{dF_X}{dx}$	Differentiate CDF to get density

2.8 KEY DISTINCTION: Mass vs. Density vs. Cumulative

This is one of the most important conceptual distinctions in probability theory.

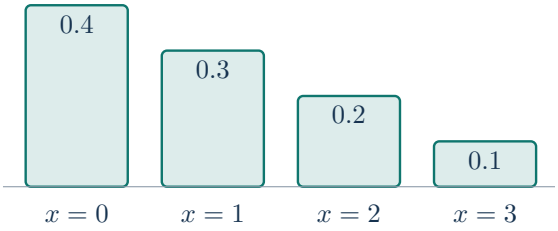
Comparison Table

Property	PMF $p_X(x)$	PDF $f_X(x)$	CDF $F_X(x)$
Applies to	Discrete RVs only	Continuous RVs only	Both
Gives directly	$P(X = x)$	Probability density at x	$P(X \leq x)$
Range of values	[0, 1]	$[0, \infty)$ — can exceed 1!	[0, 1]
Sum/Integral	$\sum p_X(x) = 1$	$\int f_X(x)dx = 1$	Starts at 0, ends at 1
Get probability	$P(X = x) = p_X(x)$	$P(a \leq X \leq b) = \int f_X(x)dx$	$P(a < X \leq b) = F(b) - F(a)$

Visual Analogy: The Warehouse

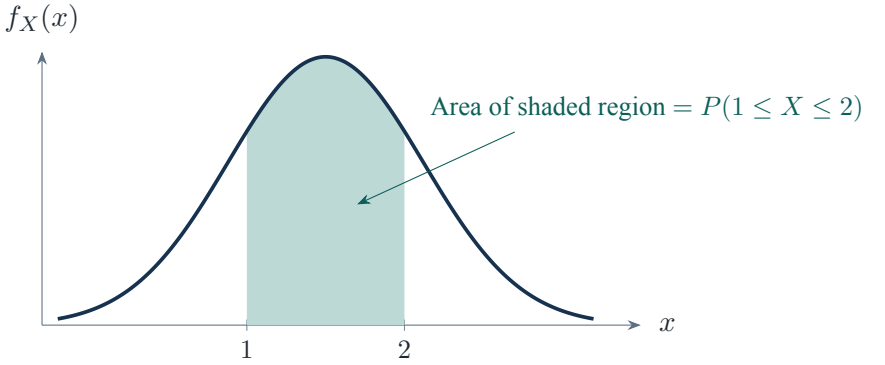
Imagine probability as "material" distributed across possible values:

Discrete (PMF): Like storing material in numbered boxes. Each box has a specific mass. The PMF tells you the mass in each box.



Total mass in all boxes $= 1$

Continuous (PDF): Like spreading material along a continuous shelf. The density tells you how thick the material is at each point. You find mass by computing area (thickness $\times$ width).



Total area under curve $= 1$

Cumulative (CDF): Like a running total — how much material have you accumulated up to this point?

Why This Matters in Engineering

Example: A noise voltage X has PDF $f_X(x) = 2$ for $x \in [0,\, 0.5]$.

- $f_X(0.3) = 2$ — this is NOT a probability! It means the density is $2\,\text{V}^{-1}$.
- $P(0.2 \le X \le 0.4) = \int_{0.2}^{0.4} 2\,dx = 2 \times 0.2 = 0.4$ — THIS is a probability.
- $P(X = 0.3) = 0$ — a single point has zero probability for continuous RVs.

Common Errors to Avoid

✘ Wrong Statement	✔ Correct Statement
"The probability at $x = 2$ is $f(2)$"	"The probability density at $x = 2$ is $f(2)$"
"$f(x)$ must be ≤ 1"	"$f(x)$ can be any non-negative value"
"$P(X = 3) = f(3)$ for continuous X"	"$P(X = 3) = 0$ for continuous X"
"PMF and PDF are the same thing"	"PMF gives mass; PDF gives density"

2.9 Transformations of Random Variables

Motivation

In engineering, we often know the distribution of one quantity but need the distribution of a **function** of that quantity:

- Know voltage $V \rightarrow$ need power $P = \frac{V^2}{R}$
- Know linear signal $X \rightarrow$ need dB value $Y = 10 \cdot \log_{10}(X)$
- Know noise $N \rightarrow$ need $|N|$ (rectified noise)

Problem Statement

Given: X with known PDF $f_X(x)$ or PMF $p_X(x)$
Find: The distribution of $Y = g(X)$

Case 1: Discrete RV Transformation

If X is discrete with PMF $p_X(x)$, and $Y = g(X)$:

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x)$$

Sum the probabilities of all x values that map to the same y.

Example: $X \in \{-2, -1, 0, 1, 2\}$ with equal probability 0.2 each.
$Y = X^2$

$Y = X^2$	X values mapping to Y	$p_Y(y)$
0	{0}	0.2
1	{-1, 1}	0.4
4	{-2, 2}	0.4

Case 2: Monotonic Transformation (Continuous)

If $Y = g(X)$ where g is strictly monotonic and differentiable:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right|$$

Or equivalently:

$$f_Y(y) = \left. \frac{f_X(x)}{|g'(x)|} \right|_{x=g^{-1}(y)}$$

Derivation via CDF method:

1. Start with CDF: $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Invert: $= P(X \leq g^{-1}(y))$ if g is increasing
3. Differentiate: $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$

Case 3: Non-Monotonic Transformation

If g is not monotonic, split the domain into intervals where g IS monotonic:

$$f_Y(y) = \sum_i \frac{f_X(x_i)}{|g'(x_i)|}$$

where x_1, x_2, … are all solutions of $g(x) = y$.

Engineering Example: Power from Voltage

A noise voltage X has PDF $f_X(x)$ (e.g., Gaussian with zero mean).
Power dissipated in a 1Ω resistor: $P = X^2$
Since $Y = X^2$ is NOT monotonic (both $+x$ and $-x$ give the same y):
For $y > 0$: $x = +\sqrt{y}$ and $x = -\sqrt{y}$ are both solutions.
$g(x) = x^2$, so $g'(x) = 2x$

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0$$

If X is zero-mean Gaussian: $f_X(x) = f_X(-x)$, so:

$$f_Y(y) = \frac{f_X(\sqrt{y})}{\sqrt{y}}, \quad y > 0$$

Engineering Example: Linear Amplification

Signal Y = aX $+b$ (amplifier with gain a and offset b).
X has PDF $f_X(x)$.
$g^{-1}(y) = \frac{y-b}{a}$, and $|\frac{d}{dy}\left[\frac{y-b}{a}\right]| = \frac{1}{|a|}$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

2.10 Practice Problems

Problem 1: Identifying RV Type

Classify each as discrete or continuous and state the range:

(a) Number of dropped calls in one hour
(b) Received signal strength (dBm)
(c) Number of bits in error in a 1500-byte Ethernet frame
(d) Time between packet arrivals
(e) Quantizer output level (3-bit quantizer)

Solution

(a) Discrete, $X \in \{0, 1, 2, \ldots\}$
(b) Continuous, $X \in (-\infty, \infty)$ practically $\sim (-120, 0)$ dBm
(c) Discrete, $X \in \{0, 1, 2, \ldots, 12000\}$
(d) Continuous, $T \in [0, \infty)$
(e) Discrete, $X \in \{0, 1, 2, 3, 4, 5, 6, 7\}$

Problem 2: PMF Properties

A random variable X has PMF: $p_X(0) = 0.1$, $p_X(1) = 0.3$, $p_X(2) = c$, $p_X(3) = 0.2$.

(a) Find c.
(b) Find $P(X \geq 2)$.
(c) Find $F_X(1.5)$.
(d) Find $P(0.5 < X \leq 2.5)$.

Solution

(a) Sum = 1: $0.1 + 0.3 + c + 0.2 = 1 \rightarrow c = 0.4$
(b) $P(X \geq 2) = p_X(2) + p_X(3) = 0.4 + 0.2 = 0.6$
(c) $F_X(1.5) = P(X \leq 1.5) = p_X(0) + p_X(1) = 0.1 + 0.3 = 0.4$
(d) $P(0.5 < X \leq 2.5) = p_X(1) + p_X(2) = 0.3 + 0.4 = 0.7$

Problem 3: PDF and CDF

A random variable has PDF:
$f_X(x) = \text{cx}$ for $0 \le x \le 4$, and 0 otherwise.

(a) Find c.
(b) Find the CDF $F_X(x)$.
(c) Find $P(1 \le X \le 3)$.
(d) Find the value x_0 such that $P(X \le x_0) = 0.5$.

Solution

(a) $\int_0^4 \text{cx}\, dx = c \cdot \left[\frac{x^2}{2}\right]_0^4 = c \cdot 8 = 1 \to c = \frac{1}{8}$

(b) $F_X(x) = \int_0^x \frac{t}{8} dt = \frac{x^2}{16}$, for $0 \le x \le 4$
$F_X(x) = 0$ for $x < 0$; $F_X(x) = 1$ for $x > 4$

(c) $P(1 \le X \le 3) = F_X(3) - F_X(1) = \dfrac{9}{16} - \dfrac{1}{16} = \dfrac{8}{16} = 0.5$

(d) $\dfrac{x_0^2}{16} = 0.5 \to x_0^2 = 8 \to x_0 = 2\sqrt{2} \approx 2.83$

Problem 4: Transformation

A sensor measures temperature X uniformly distributed on [20, 30] °C.
The output voltage is $Y = 0.1X - 2$ (linear sensor).

(a) Find the PDF of Y.
(b) Find the range of Y.
(c) Find $P(Y > 0.5)$.

Solution

(a) $X \sim \text{Uniform}[20,\ 30]$, so $f_X(x) = \frac{1}{10}$ for $20 \le x \le 30$.
$Y = 0.1X - 2$, so $a = 0.1$, $b = -2$.
$f_Y(y) = \frac{1}{|a|} \cdot f_X\left(\frac{y-b}{a}\right) = \frac{1}{0.1} \cdot \frac{1}{10} = 1$ for the valid range.

(b) When $X = 20$: $Y = 0.1(20) - 2 = 0$; When $X = 30$: $Y = 0.1(30) - 2 = 1$
Range of Y: $[0,\ 1]$. So $Y \sim \text{Uniform}[0,\ 1]$.

(c) $P(Y > 0.5) = 1 - F_Y(0.5) = 1 - 0.5 = 0.5$

Problem 5: Non-Monotonic Transformation

A voltage $X \sim \text{Uniform}[-1,\ 1]V$. The instantaneous power is $Y = X^2$.

(a) Find the CDF of Y.

(b) Find the PDF of Y.

Solution

(a) For $0 \leq y \leq 1$:

$$F_Y(y) = P(X^2 \leq y) = P\left(-\sqrt{y} \leq X \leq \sqrt{y}\right) = F_X\left(\sqrt{y}\right) - F_X\left(-\sqrt{y}\right)$$

Since $X \sim \text{Uniform}[-1,\ 1]$: $F_X(x) = \frac{x+1}{2}$

$$F_Y(y) = \frac{\sqrt{y}+1}{2} - \frac{-\sqrt{y}+1}{2} = \sqrt{y}$$

(b) $f_Y(y) = \frac{dF_Y}{dy} = \frac{1}{2\sqrt{y}}$ for $0 < y \leq 1$

2.11 Key Takeaways

1. **Random variables** convert experimental outcomes into numbers for mathematical analysis
2. **PMF** gives the actual probability for discrete RVs (bar heights sum to 1)
3. **PDF** gives probability density for continuous RVs (area gives probability, not height)
4. **CDF** works universally — gives cumulative probability $P(X \leq x)$
5. **Transformations** let us derive distributions of functions of random variables
6. The **CDF method** (find CDF first, then differentiate) is often the safest approach for transformations

Next Module: Module 3 — Important Probability Distributions